

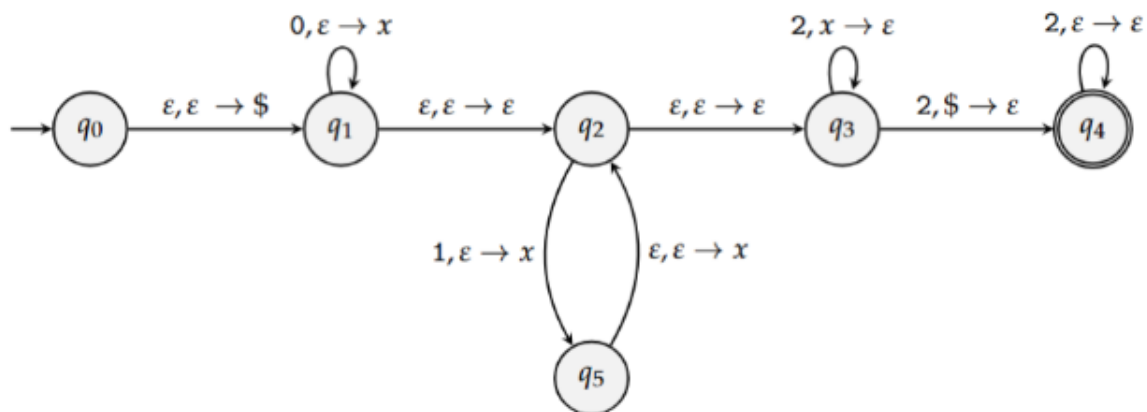
CSE331 Quiz 4	Automata and Computability, Summer 2026	
Name	ID	Section

For each statement, write True or False. For every false statement, rewrite it correctly. Each set contains exactly one true statement. (5 points)

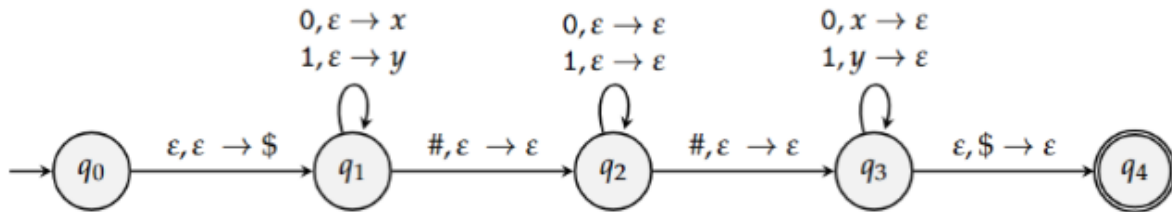
- 1) Every context-free language is also a regular language.
False. Every regular language is context-free, but not every context-free language is regular.
- 2) A nondeterministic PDA can recognize only regular languages.
False. A nondeterministic PDA can recognize all context-free languages, including nonregular ones.
- 3) A PDA's transition function always provides exactly one possible transition.
False. A nondeterministic PDA may have zero, one, or multiple possible transitions.
- 4) A language is context-free if and only if some nondeterministic PDA recognizes it.
True.
- 5) In the transition $\epsilon, \epsilon \rightarrow \$$, the PDA reads one input symbol.
False. The PDA reads no input symbol, removes nothing from the stack, and pushes \$ onto the stack

For the language given below draw the state diagram of PDA that recognizes L (4 points)

$$L = \{0^i 1^j 2^k : i, j, k \geq 0 \text{ and } k > i + 2j\}$$



$$L = \{w_1 \# w_2 \# w_1^R : w_1, w_2 \in \{0, 1\}^*\}$$

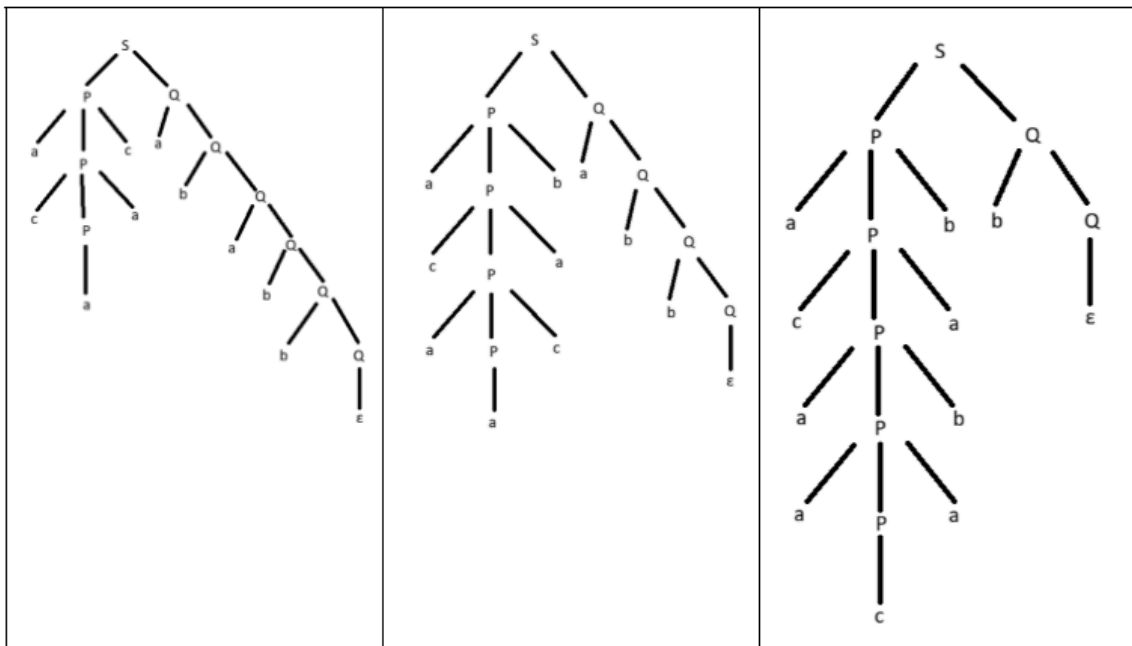


Ambiguity and Parse tree (10 points)

Consider the following context-free grammar.

$$\begin{aligned} S &\rightarrow PQ \\ P &\rightarrow aPa \mid aPb \mid aPc \mid cPa \mid bPc \mid a \mid b \mid c \\ Q &\rightarrow aQ \mid bQ \mid \epsilon \end{aligned}$$

Show that the given grammar is ambiguous by drawing two more parse trees for string 'acaacababb'. Any two parse trees.



Any two parse trees.

P produces odd-lengthed strings, and can contain a, b, c.

Q can produce strings of any length, but cannot contain c.

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- 2) A nondeterministic PDA can recognize only regular languages.
- 3) A PDA's transition function always provides exactly one possible transition.
- 4) A language is context-free if and only if some nondeterministic PDA recognizes it.
- 5) In the transition $\epsilon, \epsilon \rightarrow \$$, the PDA reads one input symbol.

For the language given below draw the state diagram of PDA that recognizes L (4 points)

$$L = \{w_1 \# w_2 \# w_1^R : w_1, w_2 \in \{0, 1\}^*\}$$

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